Noah Baculi

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EDB April 1st, 2024

Dear Recruiting Team,

I am writing to apply for the *Rust Software Engineer* position. EDB stands out to me because of the innovative techniques you utilize to tackle multifaceted database management challenges at scale and build the leading fully-managed Postgres platform. My previous experiences have honed my specialization in rewriting critical modules in Rust with increased scalability and reliability for use in the original Python, Go, or TypeScript application. My professional, academic, and leadership experiences have prepared me to excel at EDB.

In my most recent role as a Software Engineer at Carium, I developed Rust modules and Python microservice API healthcare automations and analytics. I refactored our text encoding and NLP module from Python to Rust to scale the scoring of millions of patient surveys and messages. For all our AWS lambda functions, I engineered a drop-in multithreading pattern that decreased IO bottlenecks with minimal refactoring. I pitched and developed Busca, an internal Python CLI deployment app using a file matching library I wrote in Rust that increased deployment accuracy by 15% and debugging productivity by 85%. I also designed the integration of patient data with the REST APIs of multiple electronic health records (EHR) systems to allow care team members to work natively in Carium. These team-oriented agile projects, presentations, and leadership experiences allowed me to hone the technical and communication skills that I will employ at EDB.

After graduating from Northwestern University, I stepped out of my comfort zone to become a Support Engineer at Salesforce, where I specialized in troubleshooting customer formulas, workflow rules, validation rules, and other automation code. Inspired by my experience studying for Salesforce certifications, I developed a Three.js interactive web app to serve as a resource for common Salesforce concepts and ideas. Once I started resolving customer cases, I identified opportunity for improved productivity and coded a macro program with a wxPython GUI. This utility reduced human error and increased the efficiency of case documentation tasks. My aptitude for engineering solutions and process improvements has prepared me to succeed in EDB's dynamic environment and your emphasis on learning and mentorship.

To complement my fingerstyle guitar hobby, I have been developing a tool to arrange fingerstyle guitar TABs based on the difficulty of different fingerings. The algorithms have grown in complexity as I migrated through the years from Java to TypeScript to Rust in pursuit of the best balance of performance, distribution, and developer ergonomics. The latest version implements Dijkstra's algorithm in Rust to efficiently calculate the arrangement with the least difficulty in a broadly portable mechanism via Web Assembly. I have developed the problem-solving expertise that will allow me to thrive in my responsibilities at EDB.

I believe my experience as well as my technical and leadership abilities make me a strong candidate for the *Rust Software Engineer* position. Attached is my resume that expands upon my related experience and education. I am excited to learn more about EDB and would value the opportunity to discuss this position in more detail. Please feel free to contact me at noahbaculi@gmail.com or noahbaculi.com/contact should any questions arise. Thank you in advance for your time and consideration.

Sincerely,
Noah Baculi

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SUMMARY

As a Software Engineer, I have a passion for improving efficiency and quality of life through automation technology. My core competencies include backend Python and rewriting performance-critical functions in Rust with minimal modification to the original applications. In my free time, I love being outside, reading, and playing fingerstyle guitar. I take pride in my informed, yet flexible, technical opinions that originate from my engineering background at Northwestern University and subsequent experiences solving impactful problems with high-caliber teams.

SKILLS

Python	REST APIs, Open API	Serverless - AWS Lambda
PostgreSQL	Async Microservices	Object Storage - AWS S3, Azure Blob
Rust	Regex	Data Warehouse - AWS Redshift
Go	Pandas	CI/CD - Github Actions, Jenkins
Typescript	Django, Flask	Packaging - Homebrew, PyPI

EXPERIENCE

Carium
Jan 2022 – Mar 2024

Software Engineer

- Rewrote microservice router queuing in Rust and Axum to process 57% more requests with 88% less memory usage.
- Implemented interactive Python CLI to diff and deploy distributed assets 60% faster and 20% more accurately.
- Refactored text scoring and encoding API module from Python to Rust for 78% faster survey and message scoring.
- Engineered multithreaded generic pattern to decrease the runtime of AWS Lambda IO Python API calls by 81%.
- Constructed event-driven Python API and SQL automations for a healthcare center with 50 staff to track and assign tasks dynamically with branching pathways, increasing onboarding throughput by 35% and decreasing errors by 40%.
- Developed pilot generative AI automations using AWS Bedrock and different LLMs to summarize medical histories.

Jr Software Engineer

- Created Busca, a Rust library and CLI app that increased internal file matching and searching productivity by 85%.
- Designed CI/CD pipeline for the Busca CLI to build universal binary for x86 and ARM and deploy to Homebrew.
- Developed Python bindings for Busca to deploy a Python package on MyPI, enabling standardized distribution.
- Designed health metric and medication REST API automations for a clinic with more than 23,000 patients to send patient reminders based on their metric history and alert the care team if a blood sugar reading is out of bounds.
- Integrated Carium Django APIs with two EHRs to allow care teams to use external patient data natively in Carium.
- Migrated cellular device analytics from Redshift SQL to real-time APIs, reducing end-user diagnosing delays by 86%.
- Implemented Redshift SQL materialized views, ELT and dbt pipeline, and custom Django visualization widgets for new video call feature with time tracking, billing, data export, and UI sorting and filtering in React dashboard.

Salesforce
May 2021 – Jan 2022

Software Support Engineer

- Developed SalesforceGalaxy.com, a 3D web app for learning the Salesforce platform using Vite.js and Three.js to present concise summaries of key points needed for the Salesforce Admin and App Builder certifications.
- Created help modal for SalesforceGalaxy.com that is only shown on first visit, based on the user's localStorage.
- Constructed internal macro utility with a wxPython GUI to prompt engineers with required fields and generate standardized documentation, increasing efficiency by 24%.
- Resolved over 200 client issues and exceeded KPIs for customer satisfaction, customer effort, and issue resolution.
- Debugged Salesforce formulas, validation rules, Flows, and custom API JSON responses to find root cause.
- Onboarded 2 new engineers, updated knowledge base articles, and submitted product improvement feedback.

Albireo Energy
Dec 2020 - Feb 2021

Assistant Project Manager

- Designed Python automation to parse drawing PDF documents and use regex to validate total cost values.
- Fabricated a Python wxPython GUI program to increase the speed of reviewing Request for Information (RFI) and Request for Change (RFC) documents by 60%.
- Reviewed and validated over 1,200 RFI and RFC documents to create 50 project proposal updates.
- Created visualization of contractor and construction crew availability to facilitate scheduling using NLP and Pandas.
- Updated the costs and tolerancing values using AutoCAD on over 80 engineering drawings to match requirements.

Otis Elevator
May 2020 - Jul 2020

Sales Process Intern

- Engineered bash script to automate data validation of sales leads, saving over 12 hours of repetitive work every week.
- Interacted with 260 accounts in Tulsa and Tampa territories to follow up on after-market elevator product proposals.
- Created marketing collateral and video assets for 4 elevator products to market decreased risk of viral infection.
- Collaborated across the region to create remote sales process to maximize account contacts and product education.

Trane Technologies
Jan 2019 - Aug 2019

Systems Engineering Co-Op

- Coded general-purpose automation Python scripts to increase the efficiency of repetitive computer tasks by 48%.
- Developed NLP Python program to analyze 200k rows of quality data with Tableau to identify trends in failure factors.
- Directed $43,600 instrumentation upgrade plan for 4 test loops, including ordering, calibration, and installation.
- Led team to correct $11,000 drive setting miscalculation of 60 units in production and field to increase performance.

Illinois Tool Works
Oct 2018 - Dec 2018

Engineering Intern

- Modified SolidWorks drawings to evaluate design procedures and facilitate subassembly outsourcing transitions.
- Managed for floor technician cross-training initiative, involving the creation of subassembly training strategies.
- Created new JIT ordering pipeline using historical data-driven production schedules to preemptively order parts.
- Designed new work cell and racking configuration to accommodate new diamond-wire pipe cutter product.

PROJECTS

Guitar Tab Generator – *Software Engineer*
February 2019 – Present
noahbaculi.com/guitartab

- Migrated Typescript algorithm to Rust + WASM and implemented memoization, increasing performance by 98.54%.
- Created algorithm leveraging's Dijkstra's pathfinding to calculate the optimal arrangements with the least difficulty.
- Coded pitch parser to determine the valid pitches from a string and validate for a given guitar string configuration.
- Implemented playback feature with Tone.js and rendering to allow users to preview playback of arrangements.

Aldras Automation – *Software Engineer*
October 2019 – May 2021
aldras.netlify.app

- Programmed full stack desktop SaaS application using Python to record and automate repetitive computer workflows.
- Produced website with user documentation, video tutorials, CDN download, and Stripe API payment integration.
- Managed CI/CD beta testing pipeline webhooks to integrate DevOps feedback, improving the end user experience.
- Coordinated 2 international freelance teams to design graphical assets and conduct marketing campaign A/B testing.

EDUCATION

Northwestern University – *B.S. Mechanical Engineering*
September 2017 - March 2021
Evanston, IL

GPA: 3.9/4.0 - Business Enterprise Certificate - Leadership and Service Award